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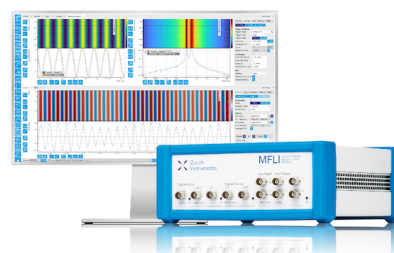
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Measurement of high viscosity with laser induced surface deformation technique

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A technique for viscosity measurement was developed based on the principle of laser-induced surface deformation. Light incident into liquids increases its momentum due to the difference in refractive index and gives the surface an upward force as a reaction. The plane surface thus swells up and deforms, and the shape is determined so that the force is balanced with the surface tension and the gravity. On sudden laser irradiation, the deformation inevitably accompanies a viscous flow and exhibits a relaxational behavior with a delay time, which gives the viscosity. Theoretical prediction of the step-response function was given that takes surface tension waves excited by the laser into consideration. Nd-yttrium-aluminum-garnet laser with 0.6 W output was focused to $\sim 200\text{ }\mu\text{m}$ beam waist and used for the pumping. The deformation process was observed sensitively with another probe laser illuminating the activated area. This system was tested with the standard liquids for viscosity ranging from 1 to 10^6 cSt . The results demonstrated the validity of this technique, though a correction for the inertia effect was needed in the range lower than 10 cSt. Further, effect of the thermal expansion by a slight optical absorption was discussed. This technique is especially useful at high viscosities since the measurement takes only a few seconds even in the specimen with 10^6 cSt . Besides the rapidity, it has a great advantage of a noncontact feature and is appropriate for measuring the liquids that strongly dislike contamination. It has also potential applications in industries, measurement of liquids isolated in a production line, for instance. © 2005 American Institute of Physics. [DOI: 10.1063/1.1839640]

I. INTRODUCTION

Liquid surface or liquid-liquid interface is acknowledged as a plane of discontinuity in physical properties that are readily observable: those mechanical characters such as density, compressibility, and viscosity as well as dielectric property are very much different between the two separated volumes. Something going across the interface feels a sudden change in the environment and may alter the motion. Laser beam, for example, feels a discontinuous index of refraction and adjusts the wavelength to the medium. This adjustment is accompanied by an increase or a decrease in the wave momentum. When the laser passes from the air down into a liquid, for example, the momentum should increase by the amount appropriate to the difference in the refractive indexes. This amount is supplied by the interface that swells up increasing its height and curvature.¹ The gravity and the surface tension both give a downward force to the convex surface, which incessantly taps the passing photons on the back. This situation being viewed from the photon side, they add their momentum by kicking back the surface and giving the upward radiation pressure to it.²⁻⁴ Therefore the surface deformation is eventually determined by a dynamic balance among the gravity of the swollen volume, Laplace force associated with the surface curvature, and the radiation pressure of the laser. The surface displacement is always toward the medium with lower refractive index and the magnitude is typically in the order of nanometer for a laser with 1 W

output. Such a small change can only be detected optically with another probe laser that illuminates the same area of the surface and picks up the convex curvature in reflection or transmission. Note that the surface is excited with a focused laser beam and has a considerable curvature at the point of nanoscale displacement.

This technique of laser-induced surface and interface deformation has been applied for measuring the interfacial tension. We have observed the very delicate interface between oil and water including surfactant molecules that diminish the interfacial tension to an extremely low level below $10\text{ }\mu\text{N/m}$.⁵ This method has a great advantage in its perfectly noncontact feature: the interface is violated by no foreign substance but for the two laser beams. There would be no fear of sample contamination that is particularly serious in surface and interface systems.

In this study, we aimed at an application of this laser-aided technique for the measurement of viscosity, that governs dynamic response of the surface to the optical perturbation. Most viscosity measurements involve the liquid into a deformation or viscous flow of macroscopic scale, and it takes a considerable time for the system to get to a stationary state ready for the observation. The measurements generally take a substantially long time, and this state of being will get more and more undesirable as the viscosity increases. If one is to measure liquids with $\eta=10^6\text{ cP}$, for instance, it would take hours to obtain a reliable value, far beyond the practicality. This problem inherent to high-viscous liquids can be solved with a microscopic deformation given to the liquid, followed by a high-sensitivity detection of the dynamic re-

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sponse. The present technique of laser induced surface deformation is sure to answer this strict requirement: the focused laser beam drives the area smaller than 200 μm into a nanometer motion. The induced viscous flow is limited in a microscopic volume and likely to settle down within a few seconds even for liquids with $\eta=10^6$ cP; and the optical probe is ready to trace the process.

II. THEORETICAL CALCULATION ON LIQUID SURFACE MOTION

In this section, we deal with the dynamic response of liquid surface to the laser irradiation, and derive the equations of time-dependent surface displacement and curvature. The calculation procedure is rather a complicated one⁶ and we give a brief outline here. Consider plane waves propagating along the x direction on the surface of liquid filling half the space $z < 0$. The fluid dynamics gives equations for particle velocity v_x and v_z in the subsurface volume.⁷ They are

$$v_x = \int_0^\infty (ikAe^{kz} - lCe^{lz})e^{i(kx+\omega t)}dk, \quad (1)$$

$$v_z = \int_0^\infty (kAe^{kz} + ikCe^{lz})e^{i(kx+\omega t)}dk, \quad (2)$$

where A and C are constants, ω and k are the angular frequency and wave number, respectively, and l is a complex wave number written with kinetic viscosity ν by

$$l = k(1 + i\omega/\nu k^2)^{1/2}. \quad (3)$$

The complex wave number includes the effect of attenuation due to the viscous loss along the depth direction. Further, the pressure p in liquid is given by

$$\frac{\partial p}{\partial t} = \omega^2 \rho \int_0^\infty Ae^{kz}e^{i(kx+\omega t)}dk - \rho g v_z, \quad (4)$$

where ρ and g are the density and gravitational acceleration, respectively. The integration with respect to k implies that the waves are not necessarily monochromatic. We take boundary conditions at $z=0$ that (i) stress component σ_{zz} normal to the surface is balanced with the sum of Laplace force p_γ of the surface tension and radiation pressure p_e of the laser; and (ii) shear component σ_{zx} equals zero. Then

$$\sigma_{zz} = -p + 2\eta \frac{\partial v_z}{\partial z} = p_\gamma + p_e, \quad (5)$$

$$\sigma_{zx} = \eta \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) = 0, \quad (6)$$

where $\eta(=\nu\rho)$ is the viscosity.

In the present study, the surface waves are excited by a modulated laser with Gaussian beam profile expressed by $\exp(-2r^2/w^2)$, w being the beam radius. Assuming a sinusoidal modulation at ω to the laser output, we get

$$p_e = \frac{4I}{cw^2\pi} \frac{(n-1)}{(n+1)} e^{-(2r^2/w^2)} e^{i\omega t} \quad (7)$$

$$= p_0 e^{-(2r^2/w^2)} e^{i\omega t}, \quad (8)$$

where r is the distance from the beam center, I_0 is the output laser intensity, n is the refractive index of the liquid, and c is the light speed in air. The laser causes a surface deformation of axial symmetry and therefore excites not plane but cylindrical waves extending over the surface. Taking this fact into account, we replace the factor $\exp(-ikx)$ in Eqs. (1), (2), and (4) with $J_0(kr)$ for the cylindrical waves, and make necessary changes including Hankel transformation. Finally, the surface displacement $\xi(r, t)$ is derived in the integration of v_z with respect to time

$$\xi(r, t) = \int_0^t v_z dt, \quad (9)$$

$$= \frac{w^2 p_0}{4\rho} \int_0^\infty \frac{k^2 e^{-(k^2 w^2/8)} J_0(kr) (e^{i\omega t} - 1)}{(i\omega + 2\nu k^2)^2 + \omega_0^2 - 4\nu^2 k^3 l} dk, \quad (10)$$

where ω_0 is given by

$$\omega_0^2 = \frac{\gamma}{\rho} k^3 + gk. \quad (11)$$

Note that the displacement brought about by the laser is superposition of waves with different wave numbers as seen in Eq. (10), and far from monochromatic in spite of the harmonic excitation. The Gaussian beam drives the surface waves in a certain wave number range, and imposes a somewhat complex feature to the waves.

Here we assume a strong attenuation of the waves and set $\omega \ll \nu k^2$. To the first-order approximation, Eq. (3) is rewritten by

$$l = \left(k^2 + \frac{i\omega}{\nu} \right)^{1/2} \simeq k \left(1 + \frac{i\omega}{2\nu k^2} \right). \quad (12)$$

Then Eq. (10) is simplified to

$$\xi(r, t) = \frac{w^2 p_0}{4\rho} \int_0^\infty \frac{k^2 e^{-(k^2 w^2/8)} J_0(kr) (1 - e^{i\omega t})}{\omega^2 + 2i\nu k^2 \omega - \omega_0^2} dk. \quad (13)$$

The denominator in the integral is converted into partial fractions and, with $\omega_0 \ll \nu k^2$, this equation is further reduced to

$$\xi(r, t) = \frac{w^2 p_0}{4\rho} \int_0^\infty \frac{e^{-(k^2 w^2/8)} J_0(kr) (1 - e^{i\omega t})}{2i\nu(\omega - i\omega_0^2/2i\nu k^2)} dk. \quad (14)$$

This assumption of strong attenuation would be quite valid in a twofold meaning: we aim at a measurement of high viscosity; and we excite waves with high k value with a sharply focused laser spot. However, we found some disagreement between the theoretical prediction and the experiment for low-viscosity liquids like water with $\nu=1$ cSt.

As described in the following section, we observe a step response of the surface to cw laser irradiation that begins at $t=0$. The response function would be given by

$$\xi^*(r, t) = \int_0^t \int_{-\infty}^{\infty} \xi(r, t) d\omega dt. \quad (15)$$

The actual laser spot size is typically smaller than $300 \mu\text{m}$ and the waves are safely considered to be in the surface tension region: The second term in Eq. (11) is therefore ignored

$$\omega_0^2 = \frac{\gamma}{\rho} k^3. \quad (16)$$

With this approximation, we finally obtain the step-response function as

$$\xi^*(r, t) = \frac{w^2 p_0}{4\rho\gamma} \int_0^{\infty} \frac{1}{k} J_0(kr) e^{-(k^2 w^2/8)} (1 - e^{-(\gamma t/2\eta)k}) dk. \quad (17)$$

This is a monotonous increasing function describing the retarded swelling up of the surface forced by the laser irradiation. Of the different optical ways to detect $\xi^*(r, t)$, we took the central curvature method in which a probe laser illuminated the active area on the surface and the far field reflection pattern was observed. The central intensity of the pattern decreases by the amount approximately proportional to the curvature and given by

$$S(t) = \left. \frac{\partial^2 \xi^*}{\partial r^2} \right|_{r=0} = \frac{w^2 p_0}{4\rho\gamma} \int_0^{\infty} k e^{-(k^2 w^2/8)} (e^{-(\gamma t/2\eta)k} - 1) dk, \quad (18)$$

$$= -\frac{2p_0}{\rho\gamma} \Gamma t e^{(\Gamma t)^2} \text{erfc}(\Gamma t), \quad (19)$$

where

$$\Gamma = \gamma/\sqrt{2\eta w}. \quad (20)$$

This is the function to be directly correlated with the experimental curve observed in this study. The marked feature of this equation is the universality in time dependence: it includes only Γt as a variable.

III. EXPERIMENT

The experimental setup used in this study is schematically shown in Fig. 1. The basic concept of the system is essentially the same as that used in our previous study.⁸ Surface deformation is excited by green output of the frequency-doubled Nd: yttrium–aluminum–garnet (YAG) laser giving 600 mW power at 532 nm. The laser beam is chopped with an acousto-optic modulator at frequencies from 0.05 Hz to 2 kHz depending on the sample viscosity, and focused onto the surface of interest by a lens with a focal length of typically 300 mm. The pump laser repeatedly picks up the surface in the phase of irradiation and releases it in the off phase. The deformation of the surface is actually retarded due to the viscosity and exhibits a kind of relaxation process. The chopping frequency is chosen so that major part of the relaxational process given by Eq. (19) appears in the phase of laser irradiation: half the period is roughly twice the char-

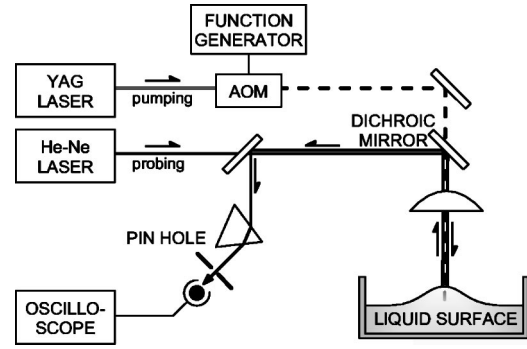


FIG. 1. Block diagram of the experimental setup for viscosity measurement. The YAG laser picks up the liquid surface, and the deformation process is detected sensitively by the He–Ne laser.

acteristic time $1/\Gamma$. Too long a period may lead to the problem of thermal expansion discussed in the following section. The measurements are made from the starting edge of each cycle on. The acousto-optic modulator operating at the ultrasonic frequency of 100 MHz yields a sufficiently sharp edge for the relaxation study.

The surface response is detected at the center of deformation, where the convex curvature gets to 0.1 m^{-1} . This value is very slight yet sufficient to be detected with the optical technique. We used the He–Ne laser with output power of 25 mW as a probe. The probe light joins the pump beam at a dichroic mirror, and is incident to the liquid surface together on the same spot. The dichroic mirror has about 100% reflectivity for red light of the probe, though almost transparent for green. For the probe laser to have sufficient sensitivity to the curvature, it should hit the center with a point that is smaller than the pump laser spot. The typical spot size of the probe laser is $100 \mu\text{m}$.

The probe light is reflected off the activated area on the surface and diverges to a large circular pattern. A photodiode with a pinhole with $150 \mu\text{m}$ diameter on its front detects the central intensity of the far-field disk. The prism works so as to refract off the green pump laser slightly left in the reflected light. The signal displayed on digital oscilloscope decreases from the value for a flat surface, and the decrease is compared with the theoretical curve of Eq. (19). We accumulated the data over 10–1000 times and determined Γ .

To check the overall performance of this system, we made a series of measurements in commercially available silicon oils used for viscosity standard. The viscosity has little temperature dependence and shows no relaxational change in the frequency range of this study. Their surface tension, density and thermal expansion coefficient are known. We prepared nine different liquids of which the viscosity ranges from $\nu=1$ to 10^6 cSt. All the measurements were made at room temperature.

IV. RESULTS AND DISCUSSION

Figure 2 shows typical examples of the step response obtained for liquids with $\nu=100$ and 1000 cSt. For a more detailed comparison with the theory, we made this particular experiment with the chopping period three to four times as long as $1/\Gamma$. The value at $t=0$ corresponds to the intensity

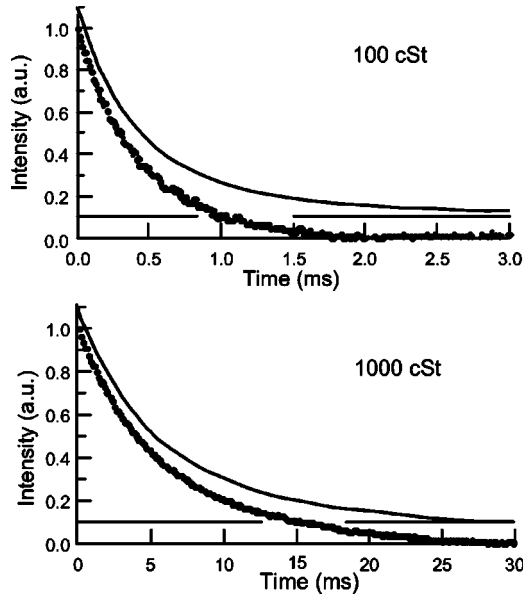


FIG. 2. Step response of the surface to the pumping laser irradiation for the liquid with $\nu=100$ cSt (upper) and $\nu=1000$ cSt (lower). The solid lines are the theoretical curves of Eq. (19), which are drawn shifted upward by 0.1 for avoiding the complete overlap with the experimental points. The theoretical curves are calculated with the known value of γ , ν for each liquid and $w=100$ μm .

reflected from the flat surface and is normalized to 1: the whole curve is biased as that. The solid lines are the theoretical curves of Eq. (19) calculated with the known values of η , γ , and $w=100$ μm . They are $\eta=96.5$ cP, $\gamma=20.9$ mN/m and $\eta=970$ cP, $\gamma=21.2$ mN/m for the samples of 100 and 1000 cSt, respectively, and hence, $1/\Gamma=0.72$ and 6.9 ms. Despite that the theoretical curves are strictly calculated with no adjustable parameter, the experimental points show the perfect agreement with them. The solid lines are shown shifted off for avoiding the complete overlap.

Figure 3 shows the relationship of kinetic viscosity ν versus the characteristic time $1/\Gamma$ on logarithmic scales, correlated with the experimental points. The thin solid line represents the prediction of Eq. (20) and is in very good agree-

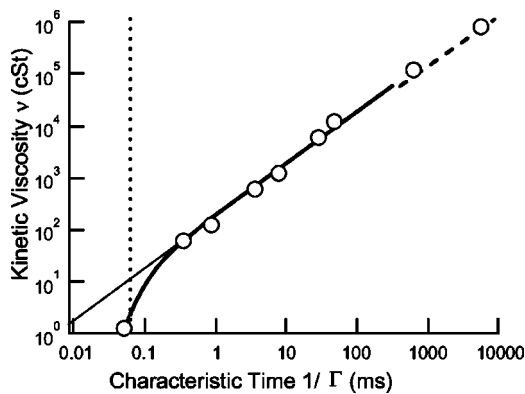


FIG. 3. Kinetic viscosity ν vs characteristic time $1/\Gamma$. The experimental values are indicated by open circles. The thin solid line exhibits theoretical prediction by Eq. (20) with $w=100$ μm , and the dashed line represents that with $w=135$ μm . The effect of inertia given by Eq. (21) is shown by the vertical dotted line. The solid curve shows Eq. (22) which includes both the effects.

ment with the data points for $\nu=50\text{--}10^4$ cSt. This result demonstrates that the present method is very useful for the measurement over this viscosity range. For liquids outside of this range, however, problems arise that may lead to errors and corrections are required. They are (i) the inertia effect of liquid motion for less viscous liquids, and (ii) the thermal expansion effect for higher viscosity liquids.

(i) *Inertia effect.* In the previous section, we made the assumption of strong attenuation and set $\omega \ll \nu k^2$. This assumption breaks for viscosities smaller than 10 cSt, however, and the lowest point in Fig. 3 is far below the theoretical line. On start of irradiation, the laser drives a mass of liquid into a sudden motion, but this motion takes an inevitable lag time for acceleration against the inertia. The earlier assumption means that this time is much shorter than the delay for dragging the volume against the viscous friction. The starting delay is determined not by viscosity but by inertia for the liquids of low viscosity. We estimate the inertia lag in the following. The laser with beam width w drives the surface wave whose wave number is given by $K=2\pi/w$. Though the surface deformation generally includes a range of wave number, this is the most dominant wave involving a mass $\sim \rho w^3$ into motion. Then the starting lag would be given roughly by the period of this wave. With the dispersion relation $\Omega^2=(\gamma/\rho)K^3$ given in Eq. (16), the time lag is estimated by

$$\frac{2\pi}{\Omega} = \left(\frac{\rho w^3}{\gamma 2\pi} \right)^{1/2}. \quad (21)$$

Considering a constant acceleration of the radiation pressure, the surface displacement begins as t^2 . The step-response curve would exhibit an inverse parabolic decrease on the start, that we actually observed in the result of $\nu=1$ cSt sample. Taking this effect of inertia lag into account, the characteristic time should be corrected as

$$\tau = \frac{1}{\Gamma} + \frac{2\pi}{\Omega} = \frac{\sqrt{2}\eta w}{\gamma} + \left(\frac{\rho w^3}{\gamma 2\pi} \right)^{1/2}. \quad (22)$$

The thick solid curve in Fig. 3 shows this theoretical prediction and the vertical line indicates the second term. Liquids with low viscosity could be measured with this curve, though the accuracy is rather limited.

(ii) *Thermal expansion effect.* All the liquids used in this study are transparent. Nevertheless they have a very slight optical absorption that causes a temperature increase in the volume of pump laser pass and around. The increase is estimated to be in the order of millidegrees and Kelvin hardly changes the viscosity. The problem arises, however, that the due thermal expansion swells the surface up and gives an additional displacement. This process takes a certain lag time, ~ 50 ms as given later, independent of the viscosity. If the chopping frequency of the pump laser is much faster than this time, the surface motion due to thermal expansion is frozen and has no influence to the response curve but a constant bias. This condition is readily satisfied when the viscosity is smaller than 10^4 cSt. In liquids more viscous than 10^5 cSt, on the other hand, the chopping period inevitably exceeds 100 ms and the response curves are interfered by the thermal expansion. This effect is unavoidable in principle,

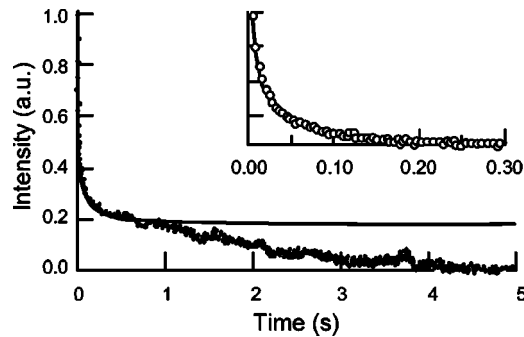


FIG. 4. The step response observed in the liquid with 10^6 cSt. The thermal expansion effect reveals itself in the initial decrease. The inset figure shows the range from 0 to 0.3 s on an expanded time scale. The solid line indicates the curve of $\sim 1/t$ fitted well to the experimental points. The thermal expansion process almost finishes within 50 ms.

but we can observe the intrinsic effect separately taking the advantage of difference in dynamics. A brief comment should be given to the reason why the thermal expansion effect is independent of the viscosity. The expansion occurs throughout the heated volume uniformly and almost concurrently with the temperature rise, and is hardly involved in viscous flow. This fact is a marked contrast to the deformation caused by the laser radiation pressure: the pump laser picks up only the surface and drags the macroscopic volume in the subsurface region against the viscosity.

We deal with this problem semiquantitatively in the following. Since the absorption occurs in the focused region, the pump laser is considered to be a rod-like heater $2w$ ($\sim 200 \mu\text{m}$) diameter and the focal depth ($\sim 6 \text{ mm}$) long. This one-dimensional shape leads to expansion in the vertical direction and pushes up the surface of free boundary condition. Though the heating begins immediately after the laser irradiation, the surrounding liquid cools the volume simultaneously via heat diffusion, and the actual course of temperature increase at the beam center (and hence, of surface swelling) is ruled by the cooling rate except the change just after the start. The two-dimensional diffusion equation suggests the time dependence of $(Dt)^{-1}$ for the surface swelling, where D is temperature diffusivity of the liquid. As the central curvature increases for this effect, the signal decreases as $\sim (Dt)^{-1}$: the observed step response would be a superposition of the term proportional to $(Dt)^{-1}$ in Eq. (19).

Figure 4 shows the curve obtained in the sample with $\nu=10^6$ cSt, which clearly exhibits the two independent processes of the thermal expansion and the deformation with viscosity: the sharp decay in the early stage followed by a slow change lasting for a few seconds. We fitted the response function of Eq. (19) to the long tail of the curve as shown and obtained $1/\Gamma=6.7 \text{ s}$, which leads to viscosity agreeing well with the known value of this liquid. The solid line in the figure indicates the curve of $1/t$ fitted to the initial decrease, which suggests that the thermal expansion process almost finishes within 50 ms. The fitting is shown on an expanded abscissa in the inset. This value agrees with the estimation roughly made with $D=2.6 \times 10^{-7} \text{ m}^2/\text{s}$ of this liquid. The experimental results show that displacement by the thermal effect is 2.5–4 times as large as the intrinsic deformation,

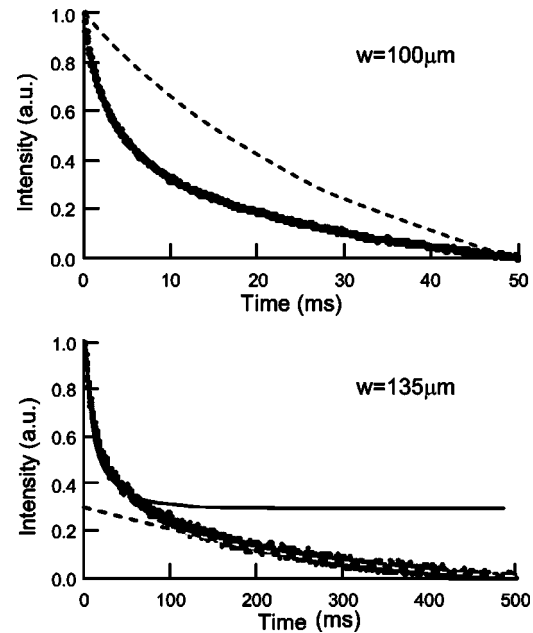


FIG. 5. The step response observed in the liquid with 10^5 cSt. The laser beam widths are $w=100 \mu\text{m}$ (upper) and $w=135 \mu\text{m}$ (lower). In the upper figure, the thermal expansion time and the characteristic time $1/\Gamma$ are close to each other and difficult to be separated. The dashed line represents the theoretical curve of Eq. (19) calculated with the known value of γ and ν , which hardly agrees with the experimental points. The two effects are separated on time scale in the lower figure. The theoretical curve of Eq. (19) is in good agreement with the experimental points after $t=0.15 \text{ s}$ and gives the accurate value of $1/\Gamma$. The solid line shows the thermal expansion process of $\sim 1/t$.

depending on the activated area. For the specimen of Fig. 5, for example, the displacement is about 20 nm. With a thermal expansion coefficient of $\alpha=9.4 \times 10^{-4} \text{ K}^{-1}$, we can estimate the temperature increase is about 11 mK. The values of D and α used in the estimation are specified for the standard specimen.

Separation of the two effects is very successful in the curve for $\nu=10^6$ cSt. The situation is not so optimistic for $\nu=10^5$ cSt, however, since the two effects are closer on time scale. Troublesome fusion of the two effects is seen in the upper curve of Fig. 5, which is by no means fitted to Eq. (19). Equation (20) tells that the intrinsic delay time $1/\Gamma$ is proportional to w , while the other is independent. Further, w has the same order of influence on magnitude of the two effects. Then we can separate them with larger w by exchanging the condenser lens, as shown in the lower curve obtained with $w=135 \mu\text{m}$.

V. SUMMARY

We applied the technique of liquid surface manipulation with laser for viscosity measurement. Dynamic response of the surface deformation to a modulated irradiation is probed with another laser illuminating the area in motion, and yields the viscosity. High accuracy is obtained over the extensive viscosity range from 10 to 10^6 cSt, though some correction is needed below 10 cSt. This technique has a great advantage of rapidity (a few seconds) in high-viscosity measurements which otherwise would require hours. This marked contrast is given by the fact that the viscous motion is limited within

a microscopic volume excited by the focused laser. The other advantage is that the liquid under study is given no mechanical disturbance except the two laser beams, and treated in a perfectly noncontact manner. This point is of particular importance for the surface with some fragile structures such as molecular layers, and has a significant meaning when the liquid strongly dislikes being polluted. Besides the purpose for physical and chemical studies, it has potential applications in the industrial fields including *in situ* measurements of liquid on production lines.

¹A. Ashkin and J. M. Dziedzic, Phys. Rev. Lett. **30**, 139 (1973).

²A. Casner and J. P. Deville, Phys. Rev. Lett. **87**, 054503 (2001).

³I. Brevik and R. Kluge, J. Opt. Soc. Am. B **16**, 976 (1999).

⁴J. Guck *et al.*, Biophys. J. **81**, 767 (2001).

⁵S. Mitani and K. Sakai, Phys. Rev. E **66**, 031604 (2002).

⁶K. Sakai, K. Tachibana, S. Mitani, and K. Takagi, J. Colloid Interface Sci. **264**, 446 (2003).

⁷V. C. Levich, *Physicochemical Hydrodynamics* (Prentice-Hall, Englewood Cliffs, NJ, 1962).

⁸K. Sakai, D. Mizuno, and K. Takagi, Phys. Rev. E **63**, 046302 (2001).